American Computer Science League

2024-2025 ● Contest 4: Shorts Problem Solutions ● Senior Division

1. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 is as follows:

$$\begin{bmatrix} 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1` & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 2 & 0 & 1 & 0 & 0 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix}$$

There are 15 pairs of vertices with at least 1 path of length 2.

A. 15

2. Graph Theory

The cycles are: ABCA, ABCDA, ABECA, ABECDA, BEB, BCDB, BCEB, BCDEB, BECDB, CEC, CDC, and CDEC. A is in 4, B is in 9, C is in 11, D is in 7 and E is in 8. Therefore, C is in the most number of cycles.

C. C

3. Digital Electronics

The digital circuit translates to: $\overline{((A + B)B) + \overline{B\,\overline{C}})\,C}$

It simplifies to:

$$\overline{((A + B)B) + \overline{B\,\overline{C}})\,C} = \overline{(AB + B + \overline{B} + \overline{\overline{C}})\,C}$$
$$= \overline{1}\,C = 0$$

Therefore, there are no ordered triples that make it TRUE.

A. 0

4. Digital Electronics

The digital circuit translates to the Boolean algebra expression:

$\overline{(\overline{A} + AB)} + \overline{\overline{(B + C)}\,\overline{C}}$

$= \overline{\overline{A} + AB + B + C + C}$

$= \overline{\overline{A} + B + C}$

$= A\,\overline{B}\,\overline{C}$

$= A\,\overline{(B + C)}$

This can be done with 2 gates: AND and NOR. The previous expression would require 4 gates: 2 NOT gates and 2 AND gates.

C. $A\,\overline{(B + C)}$

5. Assembly Language

```
X1    DC      1
X2    DC      -2
Y1    DC      1
Y2    DC      -5
      LOAD    X2
      STORE   B
      SUB     X1
      STORE   DX
      BE      STOP
      LOAD    Y2
      SUB     Y1
      STORE   DY
      DIV     DX
      STORE   M
      MULT    X1
      SUB     Y1
      MULT    =-1
      STORE   B
STOP  PRINT   B
      END
```

This program finds the y-intercept for the line between 2 points (1,1) and (-2,-5). The values of DY = -6, DX = -3, and M = 2. Therefore, B = 1 - 2(1) = -1

B. -1

6. Assembly Language

```
T         DC        0
NUM       DC        440
N         DC        =1
TOP       LOAD      T
          SUB       NUM
          BE        STOP
          LOAD      N
          ADD       =1
          STORE     N
          MULT      N
          MULT      N
          ADD       T
          STORE     T
          BU        TOP
STOP      PRINT     N
          END
```

This finds the sum of the cubes of the numbers 2^3 + 3^3 + 4^3 + 5^3 + 6^3 = 440 so the last value of N = 6.

This problem was inspired by the fact that 2^3 + 3^3 + … + 9^3 = 2024.

C. 6